

TCN Multi-Step ECG Forecasting

Ibrahim Hanafy MIT-BIH Arrhythmia Database Preprocessing: Dudukcu et al. 2023 August 2026

Experiment Setup

Model: Temporal Convolutional Network (Bai et al. 2018) — 3 residual blocks with dilated causal convolutions ($d = 1, 2, 4$), 64 filters, kernel = 3, dropout = 0.1, BatchNorm + ReLU.

Data: 6 patients (100, 103, 105, 109, 117, 124). MII lead, 100k samples each. Train/Val/Test: 40k/10k/50k. MinMax normalised (scaler fit on train only). Lookback: 10 steps (27.8 ms).

Strategies:

- Recursive** — 1-step model iterated H times
- Direct** — H independent models (H = 10 only)
- Multi-Output** — single model, H outputs (MIMO)

Horizons: 10 (27.8 ms), 20 (55.6 ms), 50 (138.9 ms)

Training: 30 epochs, batch 256, Adam ($\text{lr} = 10^{-3}$), early stopping (patience 5)

Results — Average Across 6 Patients

HORIZON	STRATEGY	RMSE	MAE	R ²	TIME (S)
10	Multi-Output	0.0375	0.0190	0.9015	41.7
10	Direct	0.0389	0.0199	0.8948	361.0
10	Recursive	0.0650	0.0308	0.6871	36.9
20	Multi-Output	0.0634	0.0311	0.7112	40.9
20	Recursive	0.1495	0.0460	-0.93	37.9
50	Multi-Output	0.0899	0.0484	0.4513	40.8
50	Recursive	831.2	1.527	-1.8×10 ⁸	43.8

Key Findings

Multi-Output wins all horizons. Best R ² at every horizon: 0.90 (H=10), 0.71 (H=20), 0.45 (H=50). Consistent, fast (~41s), no error accumulation.	Recursive collapses at H ≥ 20. Compounding prediction errors cause catastrophic divergence. Patient 124 at H=50: R ² = -1.05×10 ⁹ . Unsuitable for multi-step ECG forecasting.
Direct matches Multi-Output at H=10 but trains 9× slower (361s vs 41s). Each horizon step requires a separate model, making it impractical for larger horizons.	All strategies degrade with horizon. Even Multi-Output R ² drops from 0.90 → 0.45 as horizon grows 10 → 50. Forecasting > 139ms ahead remains challenging for ECG signals.

Where Models Failed (R² < 0.5)

PATIENT	HORIZON	STRATEGY	R ²	LIKELY CAUSE
100	10	Recursive	0.36	Error accumulation even at short horizon
100	20	Recursive	-0.02	Predictions diverge from true signal

PATIENT	HORIZON	STRATEGY	R ²	LIKELY CAUSE
100	50	Recursive	-4.0×10⁴	Catastrophic divergence
124	20	Recursive	-7.39	Predictions amplify oscillation errors
124	50	Recursive	-1.1×10⁹	Complete signal explosion
100, 105	50	Multi-Output	0.29, 0.42	Complex morphology at long horizon

Recommendation

Multi-Output (MIMO) is the recommended strategy for TCN-based multi-step ECG forecasting. It provides the best accuracy at every horizon, trains in constant time regardless of horizon length, and avoids the catastrophic error accumulation inherent to recursive approaches. For horizons beyond 50 steps, architectural improvements (larger lookback, attention mechanisms, or encoder-decoder designs) would likely be needed to push R² above 0.5.